Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **React/Node.js applications**, **Python AI systems**, scalable APIs, cloud deployments, and **mobile financial workflows** across production platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant microservices** with explicit state transitions and failure isolation, preserving recoverable background processing reliably under sustained production workloads.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured automated testing**, model switching, fallback paths, persistent results, and reviewer controls supporting reliable production workflows and troubleshooting practices.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **reproducible AI evaluation pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and controlled experiments across diverse scenarios.
- Improved closed-loop planning stability by 15% through a **simulation-based validation model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex scenarios reliably.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform**, designing reusable interfaces for customers, billing, allocation, permissions, and notifications with real-time updates across distributed locations reliably.
- Increased lead conversions by 15% through **customer and lead-management experiences**, integrating role-based access, WebSocket updates, notifications, Azure SQL, and modular services across eight operational product domains reliably.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.
- Delivered **sub-200ms interactive workflows** across 50K+ booking records, integrating pricing, coupons, payment webhooks, Firebase synchronization, and status updates while preventing conflicting reservations across concurrent user sessions reliably.

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, Bash
Frontend: React, React Native, Next.js, HTML, CSS, Responsive Design, Reusable Components, UI Development, Cross-Platform Development
Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, JSON, Microservices, JWT, Concurrent Programming
AI & Machine Learning: PyTorch, Transformers, LLMs, Reinforcement Learning, Model Evaluation, Feature Engineering, Multi-Head Attention, Vector Search, FAISS, Pinecone
Data & Real-Time Systems: MongoDB, Oracle, PostgreSQL, MySQL, Firebase, AWS RDS, Azure SQL, Caching, Query Optimization, Data Modeling, Real-Time Synchronization
Cloud & Delivery: AWS, GCP, Azure, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, Cloud Storage
Quality & Reliability: Unit Testing, Integration Testing, Automated Testing, PyTest, Jest, Selenium, Debugging, Monitoring, Error Handling, Fault Isolation, Performance Optimization, Production Reliability, Agile, Scrum

Projects

AI-Assisted Mortgage Workflow - Design Concept
- Designed an **AI-assisted financial workflow** concept with React interfaces, Python services, explainable decision signals, adaptive task routing, audit trails, human review checkpoints, privacy controls, and secure customer-data boundaries.

FinNexus | Live
- Built a **real-time fintech platform** supporting 50K+ users and 1M+ transactions with TypeScript, WebSockets, advanced filtering, MongoDB-backed data access, token-based authorization, Kubernetes deployment, and synchronized interfaces reliably.

Big-Brain | Live
- Built an **AI knowledge platform** with Python, FastAPI, PostgreSQL, vector search, FAISS, Pinecone, and cloud deployment, supporting semantic retrieval, grounded responses, and reusable application integrations reliably at scale.